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TK EfficientNet

Python · Histopathologic Cancer Detection

Notebook Input Output Logs Comments (0)

Competition Notebook	Run	Private Score	Public Score	
Histopathologic Cancer Detection	15669.8s - GPU P100	0.9334	0.9412	Version 7 of 7

GPU

Table of Contents

Histopathic Cancer Detection (HCD)

Taylor Kern

In [1]:
```
!pip install -U efficientnet -qq
```

In [2]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import pickle
import os

from sklearn.model_selection import train_test_split

import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow import keras
from tensorflow.keras.layers import *
import efficientnet.tfkeras as efn

from tensorflow.keras.preprocessing.image import ImageDataGenerator
```

Helper Functions

In [3]:
```
def merge_history(hlist):
    history = {}
    for k in hlist[0].history.keys():
        history[k] = sum([h.history[k] for h in hlist], [])
    return history

def vis_training(h, start=1):
    epoch_range = range(start, len(h['loss'])+1)
    s = slice(start-1, None)

    plt.figure(figsize=[14,4])

    n = int(len(h.keys()) / 2)

    for i in range(n):
        k = list(h.keys())[i]
        plt.subplot(1,n,i+1)
        plt.plot(epoch_range, h[k][s], label='Training')
        plt.plot(epoch_range, h['val_' + k][s], label='Validation')
        plt.xlabel('Epoch'); plt.ylabel(k); plt.title(k)
        plt.grid()
        plt.legend()

    plt.tight_layout()
    plt.show()
```

Load DataFrame

In [4]:
```
train = pd.read_csv('../input/histopathologic-cancer-detection/train_labels.csv', dtype=str)
print(train.shape)
```

```
(220025, 2)
```

In [5]:
```
train.head()
```

Out[5]:

	id	label
0	f38a6374c348f90b587e046aac6079959adf3835	0
1	c18f2d887b7ae4f6742ee445113fa1aef383ed77	1
2	755db6279dae599ebb4d39a9123cce439965282d	0
3	bc3f0c64fb968ff4a8bd33af6971ecae77c75e08	0
4	068aba587a4950175d04c680d38943fd488d6a9d	0

In [6]:
```
train.id = train.id + '.tif'
```

In [7]:
```
train.head()
```

Out[7]:

	id	label
0	f38a6374c348f90b587e046aac6079959adf3835.tif	0
1	c18f2d887b7ae4f6742ee445113fa1aef383ed77.tif	1
2	755db6279dae599ebb4d39a9123cce439965282d.tif	0
3	bc3f0c64fb968ff4a8bd33af6971ecae77c75e08.tif	0
4	068aba587a4950175d04c680d38943fd488d6a9d.tif	0

Label Distribution

In [8]:
```
(train.label.value_counts() / len(train)).to_frame().sort_index().T
```

Out[8]:

	0	1
label	0.594969	0.405031

Extract Images

In [9]:
```
train_path = "../input/histopathologic-cancer-detection/train"

sample = train.sample(n=16).reset_index()

plt.figure(figsize=(6,6))

for i, row in sample.iterrows():

    img = mpimg.imread(f'../input/histopathologic-cancer-detection/train/{row.id}')
    label = row.label

    plt.subplot(4,4,i+1)
    plt.imshow(img)
    plt.text(0, -5, f'Class {label}', color='k')

    plt.axis('off')

plt.tight_layout()
plt.show()
```